Proforma for Submitting Technical Proposal

1. General Information

S.No.	Particulars	Details to be Filled by Team
1	Hackathon Title	Smart Urban Virtual Interactive Digital Helpdesk Assistant (SUVIDHA) - 2026
2	Problem Statement ID / Theme	Design and Development of Touch-Based Kiosk Interface for Civic Utility Offices
3	Team Name	Vanquishers
4	Institution / Organisation Name	Army Institute of Technology Pune
5	Team Leader Name	Adith K P
6	Contact Email ID	adithkp03@gmail.com
7	Contact Mobile Number	9645774936
8	City / State	Pune, Maharashtra

2. Team Composition

S. No.	Name of Team Member	Role in Team	Qualification / Specialization
1	Adith K P	Frontend Developer	HTML,CSS,React.js
2	Akash Rajeev K V	Backend Developer	Python,Docker,FastAPI
3	Abindas P	Backend Developer	Node.js,PostgreSQL
4	Alan B	UI/UX Designer	Canva,Figma

3. Project Overview

3.1 Project Title:

JanSuvidha : Citizen Centric Unified Smart Kiosk for Urban Civic Services

3.2 Brief Project Description:

JanSuvidha is a citizen-centric, touch-enabled, multi-lingual smart kiosk solution aimed at improving the interface between citizens and civic utility services. The solution brings together

various departments like Electricity, Gas, Water Supply, Waste Management, and Municipal Services into a single, unified self-service interface at civic offices and public sites.

JanSuvidha allows citizens to carry out important tasks such as bill payments, service requests, grievance registration, document uploads, receipt generation, and status tracking independently, thus relieving congestion at counters and making services more efficient.

In addition to the required features, the solution includes intelligent citizen support, accessibility, offline fail-safe functionality, smart request routing, governance analytics, emergency alert broadcasting, and a secure and scalable microservices architecture. The solution is built in accordance with the Government of India IT guidelines and the Digital Personal Data Protection (DPDP) Act.

3.3 Key Features & Functionalities:

A. Mandatory Core Features (As per Problem Statement)

- Unified kiosk interface for multiple civic utility departments
- Touch-based, responsive, multilingual user interface
- Secure citizen authentication and session management
- Bill payment with receipt generation
- Service request and grievance registration
- Document upload and retrieval
- Real-time application and complaint status tracking
- Administrative dashboard for monitoring and reporting

B. Additional USPs Incorporated

1. AI-Guided Multilingual Citizen Assistant
   Provides step-by-step guidance, simplifies government terminology, and maps citizen intent to the correct service.
2. Assisted Accessibility Mode
   Includes large-font UI, audio prompts, high-contrast themes, and staff-assist triggers for elderly and differently-abled users.
3. Offline-First Fail-Safe Operation
   Ensures uninterrupted kiosk functionality during network outages with local data caching and automatic synchronization.
4. Intelligent Auto-Routing & SLA-Based Prioritization
   Automatically routes requests to the appropriate department and prioritizes them based on urgency and service-level rules.

5. QR-Based Session Continuation
   Allows citizens to continue tracking or completing requests on their mobile devices after kiosk interaction.
6. Smart Document Validation
   Performs basic validation checks on uploaded documents to reduce rejection and repeat visits.
7. Civic Analytics & Insights Dashboard
   Provides administrators with usage trends, complaint patterns, and operational insights.

4. Problem Understanding & Objective

4.1 Understanding of the Problem Statement

The rapid growth of urbanization in Indian cities has caused a substantial increase in the demand for efficient and transparent civic utility services like electricity, gas, water, waste management, and civic grievance redressal from the citizens. However, the existing service delivery model in the civic utility offices is still dependent on manual counters, paper-based systems, and departmental workflows.

This has resulted in long queues, multiple visits to the civic offices by citizens, inconsistent levels of service delivery, delayed grievance redressal, and increased administrative workload for the staff.

The citizens, especially senior citizens, new users, and less tech-savvy citizens, face difficulties in dealing with the complex processes, understanding technical jargon, and identifying the appropriate department for their service requirements. Moreover, the absence of real-time status updates causes frustration and a lack of trust in the civic systems.

From an administrative standpoint, the disintegrated service systems make it difficult to manage citizen requests efficiently, prioritize urgent citizen needs, track performance, and extract meaningful insights from citizen interactions. The instability of the network in most government offices further hampers the feasibility of fully online systems, thereby affecting service continuity.

The problem statement for JanSuvidha proposes to tackle the above challenges by conceptualizing a single, touch-enabled, and self-service kiosk that integrates all civic services into a single, easy-to-use interface. This system needs to be inclusive, multi-lingual, secure, and robust, and also aims to reduce counter dependency and increase operational efficiency. The proposed solution needs to fill the gap between citizens and civic authorities by providing independent access, real-time status updates, and data-driven governance in a feasible and implementable way that is suitable for the Indian urban context.

4.2 Objectives of the Proposed Solution

The main aim of JanSuvidha is to develop a smart kiosk solution that aims to improve the efficiency, accessibility, and transparency of civic service delivery in urban areas. The objectives of the proposed solution are as follows:

- **Unified Service Access:**
  To develop a single self-service kiosk system that provides access to various civic utility services like electricity, gas, water, waste management, and municipal services without the need to interact with different government departments.
- **Improved Citizen Experience:**
  To make service delivery easier with a touch-based guided interface that is multilingual and uses an AI-based citizen guidance system, making it easier for citizens to use, irrespective of their digital literacy.
- **Accessibility & Inclusivity:**
  To make civic services accessible to all citizens with assisted accessibility features like large font display, audio assistance, and high-contrast display for senior and differently-abled citizens.
- **Operational Efficiency:**
  To minimize the manual effort required at service counters by enabling self-service and intelligent request routing and prioritization based on service requirements.
- **Service Continuity & Reliability:**
  To ensure continuous operation of the kiosk even in the absence of network connectivity by designing the kiosk for offline usage and ensuring fail-safe functionality.
- **Transparency & Real-Time Tracking:**
  To enable real-time tracking of citizen requests, payments, and grievances, thus increasing trust and accountability in civic services.
- **Data-Driven Governance:**
  To collect and analyze data related to kiosk usage and service interactions using an administrative dashboard to enable informed decision-making and service optimization by civic authorities.
- **Security & Compliance:**
  To ensure secure authentication, protected transactions, and compliance with Government of India IT guidelines and the Digital Personal Data Protection Act.

5. Technical Design & Architecture

5.1 Solution Description

JanSuvidha is intended to be a comprehensive, touch-enabled, self-service smart kiosk solution that allows citizens to access various urban civic utility services seamlessly using a single, user-friendly interface. The solution uses a modular, microservices-based design approach to ensure scalability, security, and integrability with various departments like Electricity, Gas, Water Supply, Waste Management, and Municipal Services.

At the citizen interface level, JanSuvidha offers a responsive kiosk interface that is optimized for touch interaction. The interface allows citizens to choose their preferred language and provides step-by-step assistance for completing tasks such as bill payments, service requests, grievance registration, document uploads, and receipt generation. An AI-powered citizen assistant is also integrated into the interface to help citizens understand service options, simplify administrative jargon, and map citizen intent to the correct service flow, thus minimizing errors and confusion.

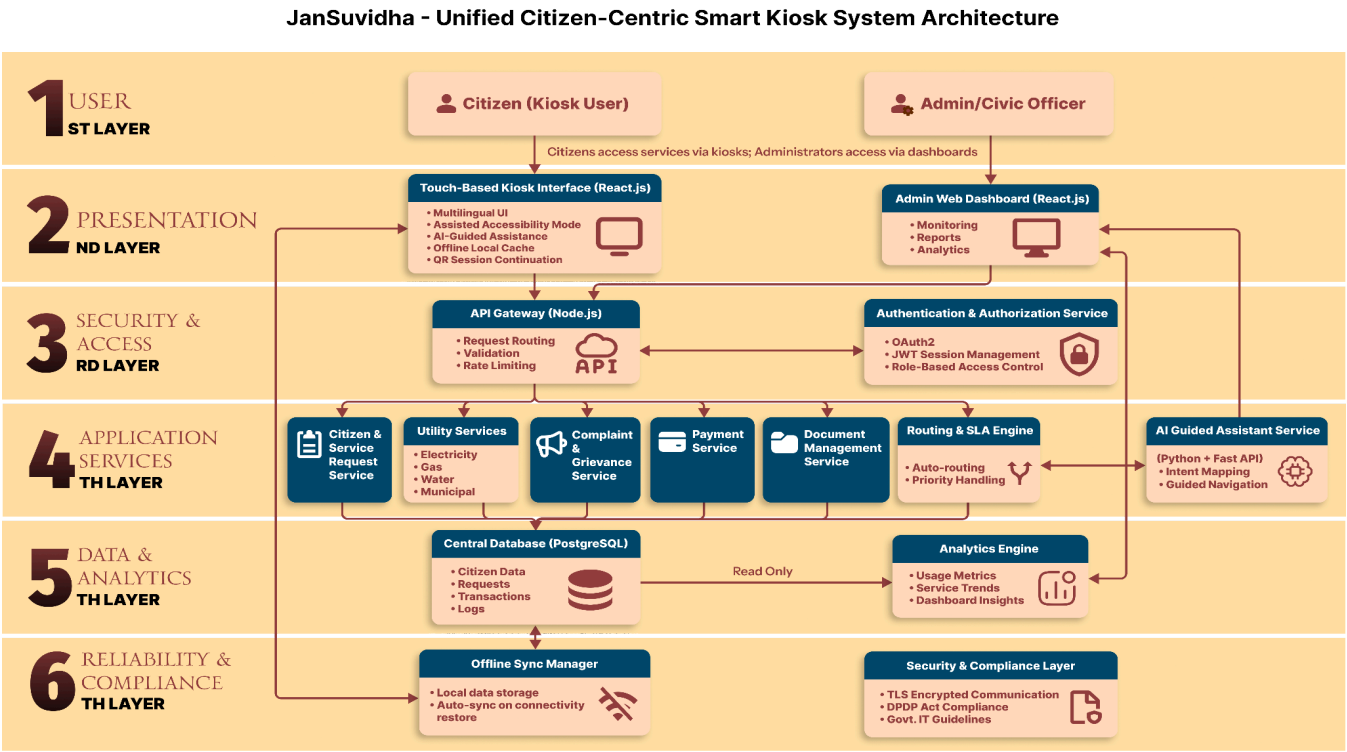
Accessibility and inclusivity are key design principles of the solution. JanSuvidha includes an assisted accessibility mode with large-font displays, audio support, and high-contrast visual themes to cater to senior citizens and individuals with disabilities. To ensure uninterrupted service delivery in real-world public office environments, the system also incorporates an offline-first fail-safe feature that allows the kiosk to continue accepting requests during network outages by securely caching data locally and synchronizing with backend services once connectivity is restored.

The backend of JanSuvidha comprises independent microservices responsible for authentication, service request management, payments, document handling, and analytics. An API gateway manages secure communication between the kiosk interface and backend services. Intelligent request routing automatically directs citizen requests to the relevant departmental services and applies priority rules based on service type and urgency, thereby improving operational efficiency and turnaround times.

JanSuvidha also supports session continuity beyond the kiosk through QR-based handoff mechanisms. Citizens receive a QR code or reference number that enables them to track request status or complete pending actions on their personal devices, reducing kiosk occupancy time. An administrative dashboard provides authorized personnel with real-time visibility into kiosk usage, service requests, and basic analytics, supporting effective monitoring, reporting, and decision-making.

Security is ensured through OAuth2-based authentication, JWT session management, encrypted communication channels, and controlled data access policies. The overall system is designed in accordance with Government of India IT standards and the Digital Personal Data Protection (DPDP) Act, ensuring data privacy, system reliability, and readiness for large-scale urban deployment.

5.2 System Architecture Diagram

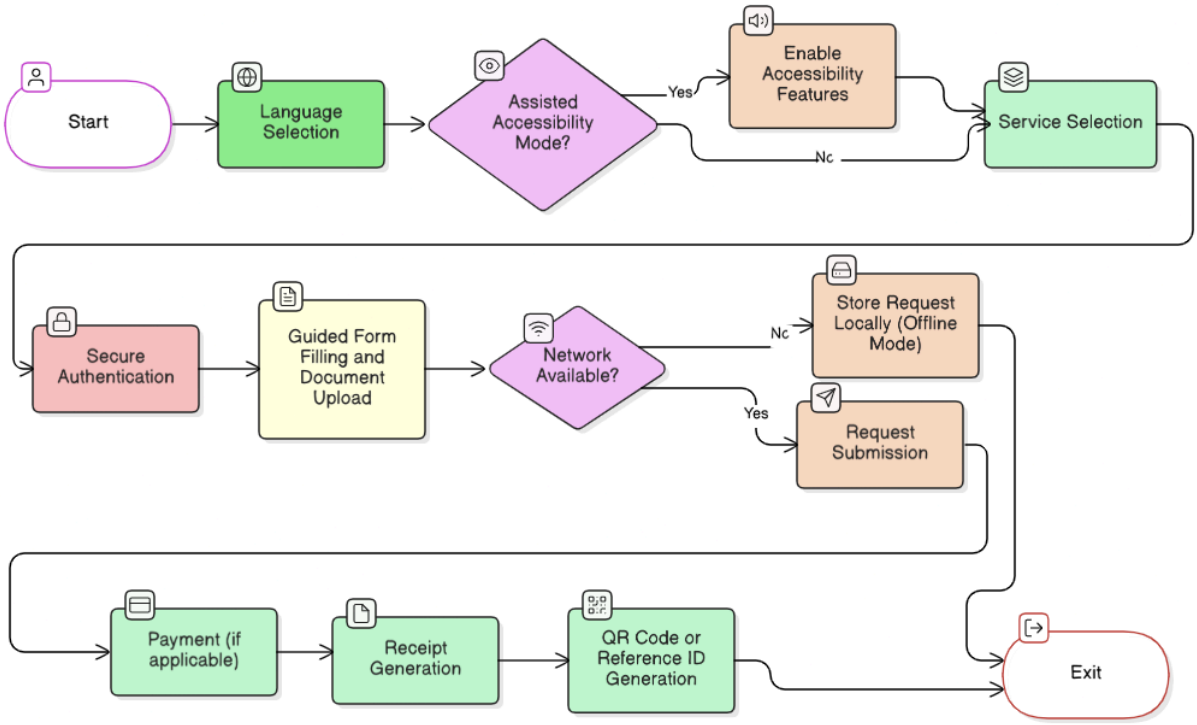


5.3 Workflow & Data Flow

The JanSuvidha system has a systematic workflow that enables secure, guided, and efficient engagement between citizens, kiosk interfaces, and backend civic service systems. The overall data flow is designed to reduce manual intervention, errors, and enable real-time visibility to both citizens and administrators.

1. **Citizen Interaction & Session Initialization**
   A citizen starts interacting with the JanSuvidha kiosk by choosing a preferred language on the touch interface. Optional assisted accessibility mode can be turned on, which enables large font displays, audio support, and high contrast displays. After choosing a service, the citizen is assisted through the process by the AI-assisted guidance module, which simplifies terms and helps the citizen choose the right service category.

2. **Authentication & Secure Session Management**
   For authenticated services, the kiosk starts the secure user authentication process through the Authentication Service. OAuth2-based authentication is done, and a JWT token is created to securely manage the user session across services. The API Gateway verifies all requests using this token to ensure authorized access.
3. **Service Request Submission**
   After authentication, the citizen submits a service request, grievance, or bill payment request through guided form workflows. The input data and uploaded files are validated at the kiosk and backend levels. The API Gateway sends the request to the corresponding backend microservice based on the service type.
4. **Intelligent Routing & Processing**
   The Routing & SLA Engine analyzes the request parameters and automatically directs the request to the concerned departmental service (Electricity, Gas, Water, or Municipal). Priority rules are also applied based on the urgency of the request and type of service, ensuring efficient processing and minimizing response time.
5. **Offline Handling & Data Synchronization**
   When the network is not available, the kiosk temporarily stores the request parameters in local, encrypted storage. The Offline Sync Manager continuously monitors the network and automatically synchronizes the pending requests with the backend services as soon as the network is restored, ensuring seamless kiosk functionality.
6. **Payment Processing & Receipt Generation**
   For bill payments, the Payment Service securely processes the transaction using the integrated payment gateway. Upon successful processing, digital receipts are generated and stored, and a confirmation message is sent to the citizen.
7. **QR-Based Session Continuation**
   After the kiosk interaction is completed, the system generates a QR code or reference number that enables citizens to check the status of their requests or complete any pending actions on their own personal devices, minimizing kiosk utilization time.
8. **Administrative Monitoring & Analytics**
   All service interactions and status updates are recorded in the central database. The Analytics Module consolidates usage information and provides actionable insights to administrators through the Admin Dashboard, facilitating monitoring, reporting, and service optimization.



6. Technical Details

6.1 **Tools & Frameworks Used:**

The JanSuvidha system is developed with the latest, open-source, and widely used technologies to make it reliable, scalable, and easy to deploy in a government infrastructure setup.

Frontend

- **Kiosk User Interface:**
  React.js with responsive and touch-friendly components
- **UI Styling & Accessibility:**
  Tailwind CSS for a consistent design, accessibility features, and responsive design
- **Offline Support:**
  IndexedDB for local data caching and offline request storage

Backend

- **API Gateway & Core Services:**
  Node.js (Express/Fastify) for business logic and service communication
- **AI-Guided Assistant Service:**
  Python with FastAPI for intent mapping and guided navigation logic

Database

- **Relational Database:**
  PostgreSQL for secure storage of user data, service requests, transactions, and logs

APIs & Communication

- **API Architecture:**
  RESTful APIs over HTTPS using JSON
- **Inter-Service Communication:**
  Secure internal REST communication via API Gateway

Authentication & Security

- **Authentication:**
  OAuth2 for secure identity management
- **Session Management:**
  JSON Web Tokens (JWT)
- **Data Security:**
  TLS-based encrypted communication

7. Implementation Details

7.1 **How does the solution work?**

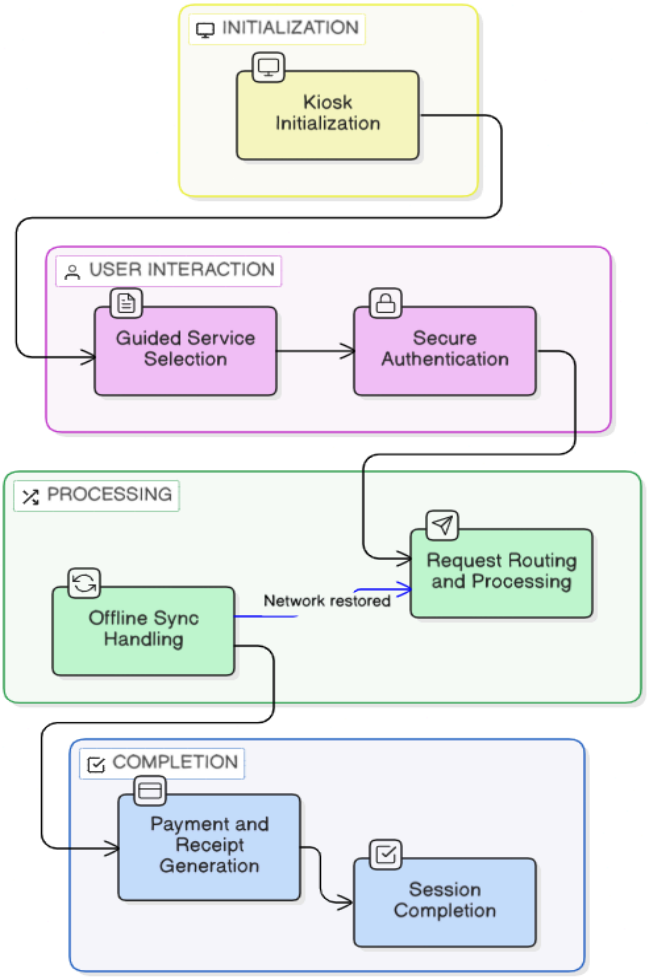
1. Kiosk Initialization & Language Selection
The JanSuvidha kiosk is initialized when a citizen interacts with the system. The citizen is first asked to choose a preferred language. If needed, the assisted accessibility feature can be turned on to support larger text, voice assistance, and high-contrast displays.

2.Guided Service Selection
The citizen chooses the desired service type (Electricity, Gas, Water, Waste Management, or Municipal Services). An AI-assisted guidance feature helps the citizen understand the available services and guides them through the correct process by simplifying government terminology.

3.Secure Authentication
For services that require identity verification, secure authentication is carried out using OAuth2-based protocols. A secure session is established using JWT tokens, ensuring safe interaction across all backend services.

4.Form Submission & Document Upload
The citizen enters the required information through a guided, step-by-step form. Supporting

documents, if needed, are uploaded via the kiosk interface. Basic validation checks ensure the correct file type and size.



5.Service Request Processing

The submitted request is routed through the API Gateway to the appropriate backend service. An intelligent routing system assigns the request to the relevant department and applies priority rules based on service type and urgency.

6.Offline Handling (If Applicable)

If network connectivity is unavailable, the kiosk securely stores the request data locally. Once connectivity is restored, the data is automatically synchronized with backend services without requiring user action.

7.Payment & Receipt Generation

For bill payment services, the system securely processes the transaction through the integrated payment module. Upon successful completion, a digital receipt is generated and displayed.

8.Session Completion & QR Continuation

After the interaction is complete, the system generates a QR code or reference number that allows the citizen to track the request status or complete pending steps later using a personal device.

9.Administrative Monitoring & Analytics

All interactions are logged and stored in the central database. Authorized administrators can monitor service activity, track requests, and review usage statistics through the admin dashboard to support oversight and decision-making.

8. Impact & Future Scope

8.1 Innovation & Uniqueness

JanSuvidha breaks away from traditional kiosk-based service delivery by going beyond simple digital replicas of forms and counters to offer a citizen-centric, intelligent, and resilient civic service platform. Unlike conventional kiosks that force users to navigate complex menus and

administrative jargon on their own, JanSuvidha integrates AI-assisted guidance that helps citizens understand services clearly and walks them through the correct workflows step by step. This approach greatly reduces confusion, errors, and reliance on staff support.

One of JanSuvidha's key differentiators is its strong emphasis on inclusivity and accessibility. The assisted accessibility mode, which includes large-font displays, audio cues, and high-contrast visuals, ensures that senior citizens, first-time users, and differently abled individuals can easily access civic services. This inclusivity-first design closely aligns with public service accessibility objectives.

JanSuvidha is also built with real-world deployment challenges in mind. Its offline-first fail-safe mechanism allows kiosks to continue operating smoothly during network disruptions, a common issue in many government offices. Requests are securely stored locally and automatically synchronized once connectivity is restored, ensuring continuity and reliability.

Another standout feature is intelligent auto-routing and prioritization of service requests. By directing requests to the appropriate departments based on predefined rules, the system reduces administrative overhead and improves response times. The QR-based session continuation feature further enhances convenience by allowing citizens to track or complete requests outside the kiosk, helping reduce congestion.

Finally, the integrated analytics dashboard equips administrators with actionable insights into service usage and trends, supporting data-driven governance. Collectively, these capabilities make JanSuvidha a practical, scalable, and deployment-ready solution tailored to the needs of Indian urban civic environments.

8.2 Future Improvements & Scalability:

JanSuvidha has been developed with a future-proof architecture that allows the system to grow and adapt alongside expanding urban requirements and ongoing technological advancements. Its modular design ensures that new services, departments, and features can be introduced smoothly without disrupting existing operations.

In upcoming phases, JanSuvidha can be strengthened through deeper integration with national and state-level digital platforms such as DigiLocker for document access, unified payment interfaces for seamless transactions, and centralized identity platforms for simplified authentication. These integrations would significantly reduce manual data entry and further improve service efficiency.

The system's analytics module can also be expanded to include advanced reporting and predictive analytics. This would help civic authorities proactively identify service bottlenecks, high-demand areas, and recurring grievance patterns, enabling better resource allocation and informed long-term urban planning.

From a scalability standpoint, JanSuvidha's microservices-based architecture supports independent scaling of individual services based on demand. The platform can be deployed

across multiple civic offices, public spaces, and cities with minimal configuration effort. Cloud-ready deployment and containerization further enable horizontal scaling and high availability.

Additionally, the platform can be extended to support mobile and web-based citizen interfaces, ensuring a seamless multi-channel experience while retaining kiosks as the primary on-site access point. Together, these future enhancements position JanSuvidha as a sustainable, scalable solution capable of supporting evolving smart city initiatives across India.

9. Scalability & Extensibility

JanSuvidha is designed to be scalable and extensible from the very beginning, allowing the system to handle growing user demand, new services, and future technological advancements without the need for major redesigns.

- **Modular Design:**
  The platform follows a modular, microservices-based architecture in which each core function—such as authentication, service management, payments, document handling, and analytics—operates as an independent service. This approach allows individual components to be updated, scaled, or replaced without impacting the rest of the system.
- **Multi-Department Integration:**
  JanSuvidha supports the integration of multiple civic utility departments within a single unified platform. New departments or services can be added by defining service-specific workflows and APIs, enabling smooth expansion beyond the initial scope of electricity, gas, water, and municipal services.
- **Independent Service Scaling:**
  Backend services can be scaled independently based on usage patterns. High-demand services such as bill payments or grievance registration can be allocated additional resources without affecting other system components.
- **Extensible API Architecture:**
  The use of standardized RESTful APIs enables easy integration with external systems, national or state digital platforms, and future mobile or web-based citizen interfaces.
- **Deployment Flexibility:**
  The system is designed for deployment across multiple kiosks, offices, and cities with minimal configuration changes, making it well suited for phased rollouts and large-scale adoption under smart city initiatives.

10. Security & Compliance

Aspect	Description
User Authentication	User authentication in JanSuvidha is implemented using OAuth2 protocols, with JSON Web Tokens (JWT) used for session management to ensure that only authenticated and authorized users can access system services.
Role-Based Access Control	Role-based access control is enforced to clearly distinguish between citizen and administrative privileges, ensuring users can access only the functions relevant to their assigned roles.
Data Privacy & Protection	Citizen data is collected and processed in compliance with the Digital Personal Data Protection (DPDP) Act. Only the minimum data required for service delivery is stored, supported by strict access control policies.
Secure Communication	All communication between the kiosk interface, backend services, and administrative dashboards is secured using TLS-encrypted channels to prevent unauthorized access, interception, or data leakage.
Secure Transactions	Payment-related activities are handled through secure transaction workflows. Transaction data is validated, securely logged, and protected against tampering to ensure accountability and integrity.
Data Storage Security	Sensitive information stored in the database is safeguarded through access controls and secure storage practices. Continuous logging is maintained to support monitoring and auditing requirements.
System Integrity & Error Handling	Input validation, request verification, and structured error-handling mechanisms are implemented across all services to prevent misuse and maintain system stability.
Compliance with Govt. IT Policies	The overall system architecture and implementation adhere to Government of India IT guidelines, cybersecurity directives, and accessibility standards applicable to smart city and civic service platforms.

11. Declaration

We hereby declare that the information provided in this technical proposal is true and correct to the best of our knowledge. The proposed solution is our original work and does not violate any intellectual property rights.

Signature: